Nebulae

Nebulae is a tangible framework for the Internet of Things. Nebulae provides networking hardware and cloud services designed and engineered to simplify the implementation of IoT solutions.

- Provides zero-configuration connection of sensors, devices and equipment.
- Provides certified, low-power, long-range networking with a guaranteed quality of service.
- Nebulae SDKs speed the development of cloud and device applications.
- Zero - collision, low power packet exchange allowing high rate for connection.
- Enhanced power management

NebuLink gateway offers a key ingredients for enabling the connectivity of industrial devices and other systems. It integrates technologies and protocols such as Modbus, SCADA & RS485 for networking, embedded control, sensing and easy manageability on which applications software can run.

Communication

Nebulae provides solution for wide range communication protocols such as WiFi, BLE, GSM (2G,3G), Ethernet, PowerLine, 6LoWPAN, 802.15.4, ZigBee.

WSN

NebuLink network has time synchronization ensuring accurate network timing that is crucial to IoT network performance.

NebuLink network is self configuring and self healing. Self healing network detects and reroutes broken path immediately. Nebulae supports Mesh, Star, Tree and Hybrid network topology.

Security

Nebulae provides Data Confidentiality, Data Integrity and Data Freshness.

Delivers a highly secure and scalable network with 3-tier encryption using the AES 128 bit standard.

Nebulae Cloud Messaging Services provides secure, scalable, real-time communication with IoT and mobile devices.

Interoperability

Nebulae provides interoperability between NebuLink devices running different stacks such as 802.15.4 and ZigBee.

NebuLink devices can communicate with other networks too, with help of NebuLink gateway working on OSGI and eclipse IoT standards.

Firmware Update

Firmware Over The Air (FOTA) is a cost-effective, reliable and secure method for updating connected devices. Secure Firmware Update with dynamic bandwidth support.

NebuLink device decides FOTA speed based on WSN traffic that does not affect performance of other NebuLink device during FOTA and manages version control on thousands of devices simultaneously.

Sensors

NebuLink devices are equipped with interface for integration of various sensors not limited to, Smoke Sensor, IR Sensor, Ultrasonic Sensor, Sound Sensor, Accelerometer Sensor, Temperature and Humidity Sensor, RF Switch/Plug, Pressure Sensor, CO_2 Sensor.

Provides real-time reading from sensors to application.